

Sandeep Sangem

Sr. Data Engineer



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PROFESSIONAL SUMMARY

- With over **7+ years of professional IT experience**, strong expertise has been built in Big Data engineering with a primary focus on **Hadoop and Spark**, supported by deep knowledge of Hadoop architecture components such as **HDFS (NameNode, DataNode), TaskTracker, JobTracker, and MapReduce programming**.
- Extensive experience includes the **development and deployment of applications** using Hadoop ecosystem components such as **MapReduce (MR1), YARN (MR2), HDFS, Hive, Pig, HBase, Flume, Sqoop, Spark (Streaming, Spark SQL), Storm, Kafka, Oozie, Zookeeper, and Parquet**, along with analytics on distributed Hadoop clusters using **Apache Spark, Impala, and Scala**.
- Advanced functionality in **Hive and Pig** has been extended through the creation of **custom UDFs and MapReduce scripts** using **Java and Python**, along with hands-on experience in setting up and scaling clusters on **Amazon EC2 and S3**, including automation of cluster configurations.
- In real-time data processing environments, **Spark Streaming jobs** have been developed in **Scala**, leveraging **RDDs, PySpark, and Spark Shell**, with cross-functional collaboration supporting reporting and analytics delivery using tools such as **Tableau**.
- Strong proficiency in **Scala and Java programming** has enabled the implementation of **OLAP multi-dimensional cube functionality** using **Azure SQL Data Warehouse**, along with data import and export operations between **HDFS and Hive using Sqoop**, and hands-on exposure to **NoSQL databases such as HBase and Cassandra**.
- Broad experience spans **structured, semi-structured, and unstructured data**, including the implementation of **complex MapReduce programs** using design patterns and best practices across large-scale data environments.
- Solid understanding of **data architecture principles**, covering **data ingestion pipeline design, Hadoop information architecture, data modeling, data mining, machine learning concepts, and advanced data processing techniques**.
- Hands-on expertise with major Big Data technologies such as **Apache Kafka, Apache Spark, Zookeeper, and Avro**, complemented by strong knowledge of **AWS services**, including **EC2, S3, Athena, RDS, VPC, IAM, Elastic Load Balancing, Lambda, Redshift, Auto Scaling, CloudFront, and CloudWatch**.
- Proven experience in implementing **unified data platforms** using **Kafka producers and consumers**, along with pre-processing and streaming logic using **Storm topologies**, and working with multiple Hadoop distributions such as **Cloudera and Hortonworks**.
- Recognized as a **strong team contributor and quick learner**, with effective communication, organizational skills, and attention to detail, actively participating across the **entire SDLC lifecycle**, including requirements gathering, design, development, testing, and production deployment.
- Strong experience in **designing end-to-end data ingestion frameworks**, handling batch and streaming data sources while ensuring reliability and scalability.
- Proven ability to **optimize Spark and Hive workloads** through effective partitioning, bucketing, caching strategies, and query tuning techniques.
- Extensive hands-on experience in **data quality validation and reconciliation**, including source-to-target count checks, schema validation, duplicate detection, and threshold-based controls.
- Experience supporting **production data pipelines**, including monitoring job executions, analyzing logs, troubleshooting failures, and implementing long-term fixes.
- Skilled in **performance tuning of large-scale data pipelines**, reducing execution time and improving resource utilization across distributed systems.
- Practical experience working in **regulated and compliance-driven environments**, ensuring data governance, access control, and audit requirements are met.
- Strong understanding of **fault tolerance and recovery mechanisms** in distributed data systems, including restartability, backfills, and rerun strategies.
- Hands-on experience in **CI/CD automation for data pipelines**, enabling consistent and reliable deployments across development, testing, and production environments.
- Experience collaborating with **upstream source teams and downstream analytics teams** to align data definitions, resolve discrepancies, and ensure data usability.
- Experience building **metadata-driven frameworks**, allowing pipelines to scale across multiple datasets with minimal code changes.

TECHNICAL SKILLS

Cloud Technologies	Azure Data Factory, Azure Blob Storage, Azure Data Lake Storage (ADLS) Gen 2, Azure Data Bricks, Unity Catalog, Azure Synapse Analytics, Azure Stream Analytics, Azure Functions, Azure Event Hubs, Azure SQL Database, Azure Active Directory, Amazon S3, Redshift, AWS Glue, Athena, AWS Lambda Amazon EMR Amazon Redshift clusters,
Hadoop Technologies	Apache Hadoop, Cloud era Hadoop Distribution (HDFS and Map Reduce) Technologies HDFS, YARN, MapReduce, Hive, Pig, Sqoop, Flume, Spark, Kafka, Zookeeper, and Oozie
Languages	Python, SQL, PL/SQL, Pyspark, Linux, Unix Shell Scripting
Modeling Techniques	Data Modeling, Star Schema Modeling, Snowflake Schema Modeling, FACT and Dimensions Tables, Slowly Changing Dimensions (SCD), Change Data Capture (CDC).
Databases	MS SQL Server, Oracle 11g/12c, Azure SQL DB, MongoDB, Teradata, My SQL, Oracle, PostgreSQL.
Business intelligence	Power BI, Qlik Sense, Spotfire,. AWS CloudWatch
Scheduling Tools	Oozie, Apache Airflow.
Version Control and CI/CD Platform	Git, GitHub, GitLab CI/CD, Maven, Jenkins, Docker, Kubernetes (Desktop client)

EDUCATION BACKGROUND

Master of Science: Computer Science

- Chicago State University – Chicago IL 2022 -2023
- GPA – 4/4
- Expect research on Data Engineering with AI Technologies.
- Honored Graduate University student of the year 2023 from field of Science and Technology.

Bachelor of Technology: Electrical and electronics engineering

- VNR Vignana Jyothi Institute of engineering and technology – Hyderabad INDIA 2011-2015

PROFESSIONAL EXPERIENCE

Client: BANK OF AMERICA, NJ
Role: Senior Hadoop Developer
Responsibilities:

Aug 2024 – Till Date

- Actively contributed to the development of real-time streaming pipelines using Apache Spark / Spark Streaming and Kafka, including Spark performance tuning, parallel processing optimization, and configuration tuning.
- Migrated MapReduce jobs to Spark, leveraging Spark SQL and DataFrame APIs on Hadoop YARN, and delivered curated datasets to BI teams for reporting and analytics.
- Developed Spark applications using Scala and Spark SQL, including frameworks for filtering invalid records, ingesting data from RDBMS sources, and exporting transformed data to Cassandra based on business requirements.

- Extracted data from Teradata into HDFS, databases, and dashboards using Spark Streaming, and developed optimized SQL queries and ETL processes within Teradata.
- Implemented and optimized Teradata utilities such as BTEQ, FastLoad, MultiLoad, and FastExport, leading end-to-end Hadoop-to-Teradata migrations, including schema conversion, datatype mapping, and high-volume ETL redesign.
- Improved Teradata performance through parallel ingestion, workload management, columnar compression, and efficient data modeling for scalable analytics.
- Designed and delivered Power BI and Tableau reports, migrated HQL logic to PySpark, optimized Hive workloads by migrating queries to Impala, and built entity resolution pipelines using PySpark and SQL.
- Implemented fuzzy matching, match/merge workflows, survivorship rules, and golden-record generation to support master data management and regulatory-ready processing.
- Built data validation, exception-handling, and audit frameworks across distributed platforms to ensure data quality, lineage, and compliance.
- Integrated log data using Apache Flume, worked with HBase and NoSQL systems, developed HiveQL scripts for aggregation and denormalization, and implemented MapReduce counters for metrics collection.
- Worked with diverse file formats (ORC, Parquet, Avro) and compression codecs (GZIP, SNAPPY, LZO), and loaded golden datasets into Apache Solr using Morphline.
- Developed Kafka-based applications to monitor consumer lag, implemented Spark Streaming transformations for near-real-time ingestion from Kafka, and persisted processed data into Cassandra.
- Managed Azure infrastructure, integrated Azure Log Analytics, and implemented Infrastructure as Code using Terraform, with additional exposure to AWS and GCP for analytics and ML workloads.
- Applied Spring and Hibernate for data persistence, leveraged Spark MLlib for predictive analytics use cases, and developed Python and PySpark solutions for large-scale data extraction and processing.
- Performed performance analysis and tuning of Spark batch and streaming jobs, while actively contributing within Agile/Scrum environments, including sprint planning, daily stand-ups, and design discussions.

Client: AT&T Inc
Role: Azure Data Engineer
Responsibilities:

Feb 2024 to Nov 2024

- Designed and developed medium to large-scale BI and data engineering solutions on Azure, leveraging Azure Data Factory, Azure Data Lake Gen2, Azure Databricks, Azure Synapse Analytics, and Azure SQL Data Warehouse.
- Partnered closely with business analysts and data engineers to deliver end-to-end solutions, including building and maintaining a centralized Azure Data Lake supporting multi-terabyte datasets.
- Built robust data ingestion pipelines connecting to source systems such as Oracle, SQL Server, and Teradata using Self-hosted Integration Runtime.
- Developed and optimized PySpark and Spark SQL workflows on Azure Databricks, converting Hive SQL scripts and enabling advanced analytics across batch and streaming workloads.
- Automated and optimized Databricks workflows, reducing operational costs while improving performance for both real-time and batch transaction processing.
- Designed and implemented event-driven architectures using Azure Event Hubs and Azure Stream Analytics to ingest, cleanse, enrich, and aggregate high-volume streaming data.
- Implemented data models, partitioning strategies, and indexing in Azure Synapse Analytics to improve query performance and resource efficiency.
- Established Delta Lake architecture in Databricks and integrated curated datasets into Azure Synapse for live Power BI dashboards.
- Developed scalable pipelines to ingest, transform, and deliver structured and unstructured data using PySpark, SQL, and cloud data warehouse technologies.
- Implemented role-based access control and secure secret management using Azure Key Vault integrated with Microsoft Entra ID (Azure AD).
- Led data governance initiatives, implementing Microsoft Purview and Unity Catalog for metadata management, access control, and regulatory compliance.
- Ensured adherence to data governance and compliance frameworks (including HIPAA), implementing security, privacy controls, and audit-ready pipelines.
- Designed and maintained cost-effective cloud infrastructure, optimizing PaaS and IaaS resources and implementing monitoring, logging, and alerting using Azure Monitor.
- Worked with Azure Functions, Logic Apps, and Function Apps for event-based and scheduled orchestration, including Power BI extract automation and SharePoint integrations.
- Utilized Azure Data Explorer for real-time analytics, enabling fast querying and visualization of streaming datasets.
- Deployed containerized data processing workloads using Kubernetes, supporting scalable and resilient data services.
- Designed and supported PostgreSQL databases, ensuring reliable data storage and query performance.

- Built data pipelines supporting data science and advanced analytics workloads, with a strong focus on data quality and transaction monitoring.
- Managed source code using Git, GitHub, and Azure DevOps, actively participating in Agile ceremonies including daily stand-ups, sprint reviews, and PI planning.
- Monitored production pipelines, resolved issues proactively, and implemented preventive measures to improve system stability and reliability.

Tata Consultancy Services

Client: Bank of America

Sept 2017 to Aug 2022

Role: Software Developer

Responsibilities:

- Built and maintained Python-based startup and shutdown automation for Hadoop and Spark services, ensuring smooth cluster operations and reducing manual intervention.
- Designed and implemented near-real-time data processing using Spark Streaming (Scala), while leading the migration of legacy MapReduce workloads to Spark for improved performance and maintainability.
- Developed and managed Apache NiFi data flows to reliably integrate data from Mainframe systems, HBase, and MongoDB into the Hadoop ecosystem.
- Implemented robust ETL workflows on Hadoop YARN using Sqoop to import and export data between Teradata, DB2, and HDFS.
- Configured and managed Zookeeper to enable coordinated access across Hive services, and implemented Slowly Changing Dimension (SCD Type 2) auditing to support historical data tracking.
- Optimized Hive workloads through effective partitioning, bucketing, and distributed cache usage, significantly improving query performance and lowering processing costs.
- Built streaming ingestion pipelines using Kafka, including Python-based publisher APIs, and standardized data storage using Avro and Parquet formats with compression.
- Automated build, deployment, and release processes using Jenkins, and implemented infrastructure automation using Terraform and Kubernetes to improve scalability and consistency.

Tata Consultancy services

Client: BSNL

Mar 2016 – Aug 2017

Role: Software Developer

Responsibilities:

- Designed and implemented star and snowflake schemas, building scalable dimensional data models to support enterprise reporting and analytics.
- Developed and maintained ETL workflows using Informatica and SSIS, integrating data from relational databases, flat files, and external APIs into the data warehouse.
- Implemented data quality checks, validation rules, and metadata management to ensure accuracy, traceability, and reliable data lineage.
- Built and supported SSRS reports and dashboards, delivering business-critical insights and KPIs to stakeholders.
- Managed day-to-day data warehouse operations, including job monitoring, performance tuning, backups, and routine maintenance to ensure system stability.
- Automated deployments and supported containerized services using Kubernetes and Helm, improving consistency and reducing operational overhead.

Client: Conquer Technologies Inc

Dec 2014 – Mar 2016

Role: Software Quality Assurance Engineer

Responsibilities:

- Worked with multiple technical teams, including the Data team, to initiate, design, manage, and deliver key QA and Data Quality components and projects.
- Involved in understanding the QA requirements throughout each phase of a project.
- Implemented QA controls, data reporting, and data quality foundations.
- Assisted with QA processes, conversion of project specifications into test plans, and manage the QA execution activities from start to finish.
- experience in work management tools (Jira, DevOps, Confluence, Asana, etc.) along with data governance and data quality management tools (Collibra, Monte Carlo, etc.).
- hands-on QA within a Data and Analytics organization.